



Time Series Analysis

Chapter 13: Speculative bubbles: LPPL models



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Acronyms used in this material

- ▣ **ADF**: Augmented Dickey–Fuller (test)
- ▣ **AI**: Artificial Intelligence
- ▣ **AR**: AutoRegressive (model)
- ▣ **BET**: Bucharest Exchange Trading (index)
- ▣ **BET-TR**: BET Total Return (index)
- ▣ **BSADF**: Backward Supremum Augmented Dickey–Fuller (statistic)
- ▣ **EODHD**: EOD Historical Data (market-data provider)
- ▣ **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **GSADF**: Generalised Supremum Augmented Dickey–Fuller (test)
- ▣ **JLS**: Johansen–Ledoit–Sornette (model)
- ▣ **LPPL**: Log-Periodic Power Law
- ▣ **LPPLS**: Log-Periodic Power Law Singularity (model)
- ▣ **OLS**: Ordinary Least Squares
- ▣ **SADF**: Supremum Augmented Dickey–Fuller (test)
- ▣ **SE**: Standard Error
- ▣ **SSR**: Sum of Squared Residuals

The question of the chapter

- **Question:** can a speculative bubble be recognised in a price series before it bursts?
 - ▶ a **speculative bubble**: a price that rises far above the value justified by the expected cash flows of the asset, sustained by the expectation of selling it later at a higher price
 - ▶ a **crash**: a large fall of the price in a short time (here: at least 20% from the peak)
- **Route** of the chapter
 - ▶ bubbles in history and the statistical signatures of a bubble
 - ▶ rational bubbles and explosive roots: the right-tailed unit-root tests of Phillips, Shi and Yu (link with [Chapter 3](#))
 - ▶ the LPPL model of Johansen, Ledoit and Sornette: super-exponential growth, log-periodic oscillations, the critical time
 - ▶ estimation, confidence from many windows and an honest evaluation of crash prediction

Self-study guide

- This chapter is for **self-study**
 - ▶ there is no seminar
 - ▶ each section ends with a recap
 - ▶ read a section, then redo its worked example on paper before you look at the solution
 - ▶ after each chart, read the interpretation slide and compare it with your own reading of the chart
- Run the lecture notebook: every chart and number of the slides is produced there
 - ▶ change one setting (the window, the filter, the threshold) and see how the conclusion changes
- Check yourself with the self-assessment at the end and with the quiz of [Chapter 13](#) on the course site
- Prerequisites: log returns and log prices ([Chapter 1](#)), the Dickey–Fuller test ([Chapter 3](#)), OLS regression

Learning outcomes

- ▣ Describe a bubble by its fundamental value and by its statistical signatures
- ▣ Explain why a rational bubble must grow at an explosive rate and test for explosive roots with SADF and GSADF
- ▣ Write the LPPL equation and interpret each of its seven parameters
- ▣ Estimate LPPL with the two-step method of Filimonov and Sornette and check the filter conditions
- ▣ Build a confidence indicator from many windows and evaluate its alarms honestly, with false alarms and base rates

Reading and tools

- Textbook background: [Huang and Petukhina \(2022\)](#) (unit roots and nonlinear least squares, the tools of this chapter)
 - ▶ the LPPL model: [Johansen, Ledoit and Sornette \(2000\)](#); [Sornette \(2017\)](#); calibration: [Filimonov and Sornette \(2013\)](#)
 - ▶ explosive roots: [Phillips, Wu and Yu \(2011\)](#); [Phillips, Shi and Yu \(2015a\)](#); survey: [Phillips and Shi \(2020\)](#)
 - ▶ history of bubbles: [Kindleberger and Aliber \(2005\)](#); [Shiller \(2015\)](#); [Garber \(1990\)](#)
- Python Quantlets of this chapter: [Quantlets/Ch_13](#)
 - ▶ the SADF/GSADF statistics and the LPPL calibration are written out in the code (no special package)
- Lecture notebook: [open in Google Colab](#)
- Data: daily closes of the S&P 500, Nasdaq 100, BET, Shanghai Composite and Bitcoin (EODHD)

Four centuries of manias



Flora's wagon of fools, c. 1637

Painting: [Hendrik Gerritsz Pot \(c. 1637\)](#); public domain;

[Wikimedia Commons](#)

- **Tulip mania**, Netherlands, 1636–1637
 - ▶ contracts on rare bulbs traded at extreme prices, then collapsed in February 1637
 - ▶ [Garber \(1990\)](#): part of the story is a myth; rare bulbs were expensive for good reasons, common bulbs show the mania
- **South Sea** (1720), **Wall Street** (1929), **Japan** (1989), **dot-com** (2000), **housing** (2007), **crypto** (2017, 2021)
- The common pattern [Kindleberger and Aliber \(2005\)](#): a new story, easy credit, rising prices that attract new buyers, then panic

1929 and 1987



Crowd outside the New York Stock Exchange, 29
October 1929

Photo: US government (1929); public domain; Wikimedia
Commons



Amsterdam Stock Exchange, 21 October 1987

Photo: Bart Molendijk / Anefo, Nationaal Archief (1987);
CC0; Wikimedia Commons

□ October 1929

- ▶ the end of the boom of the 1920s and the start of the Great Depression

□ 19 October 1987 (Black Monday)

- ▶ the largest one-day fall of the Dow Jones index, without any major news that day
 - ▶ [Sornette, Johansen and Bouchaud \(1996\)](#) found accelerating, log-periodic oscillations in the years before 1987; this observation started the LPPL literature
 - ▶ the daily data of this course start in 1990: we study 1987 through the literature and later bubbles with our own data
- A crash without news suggests an **endogenous** cause: the market itself became fragile

Fundamental value and bubble

□ Fundamental value F_t

- ▶ the present value of the expected future cash flows (dividends D), discounted at the rate r
- ▶
$$F_t = \sum_{k=1}^{\infty} \frac{E_t[D_{t+k}]}{(1+r)^k}$$
- ▶ D_{t+k} : the dividend paid k periods from now; $E_t[\cdot]$: the expectation given the information available at t ; r : the discount rate per period

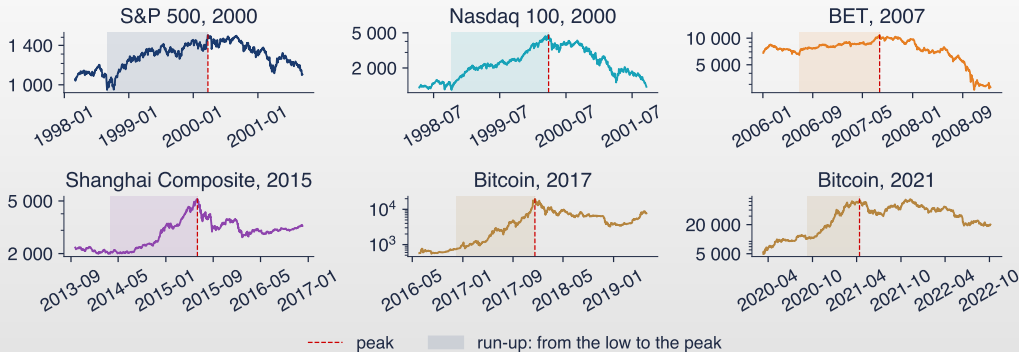
□ Bubble component

- ▶ $B_t = P_t - F_t$, the part of the price P_t not explained by fundamentals
- ▶ F_t is not observed: we can only test the **implications** of a bubble in the price path

□ Two views

- ▶ efficient markets: prices reflect information; “bubbles” are visible only with hindsight [Fama \(2014\)](#)
- ▶ behavioural finance: herding and extrapolation push prices away from value [Shiller \(2015\)](#); large run-ups do raise the probability of a crash [Greenwood, Shleifer and You \(2019\)](#)

Six run-ups and crashes in the course data



- Daily closes, log scale, from six months before the low of the run-up to eighteen months after the peak; shaded: the run-up from the low to the peak

Interpreting the six episodes

Episode	Low	Peak	Rise	Growth p.a.	Fall in 1 year
S&P 500, 2000	31 August 1998	24 March 2000	60%	30%	27%
Nasdaq 100, 2000	8 October 1998	27 March 2000	317%	97%	66%
BET, 2007	27 June 2006	24 July 2007	68%	48%	50%
Shanghai Composite, 2015	20 March 2014	12 June 2015	159%	77%	49%
Bitcoin, 2017	1 December 2016	16 December 2017	2476%	312%	83%
Bitcoin, 2021	8 September 2020	13 April 2021	527%	309%	53%

- Growth p.a.: $\ln(P_{\text{peak}}/P_{\text{low}})$ divided by the length of the run-up in years; fall: the lowest close in the year after the peak, relative to the peak
- Bitcoin multiplied its price 26 times in about a year; the S&P 500 rose much less: not every peak is a bubble of the same strength

Recap: Bubbles in history

- A bubble is a price above the fundamental value F_t ; F_t is not observed
- Six episodes in our data: run-ups of 60% to 2476%, followed by falls of 27% to 83% within a year
- Signatures: explosive and super-exponential growth, accelerating oscillations, then an end

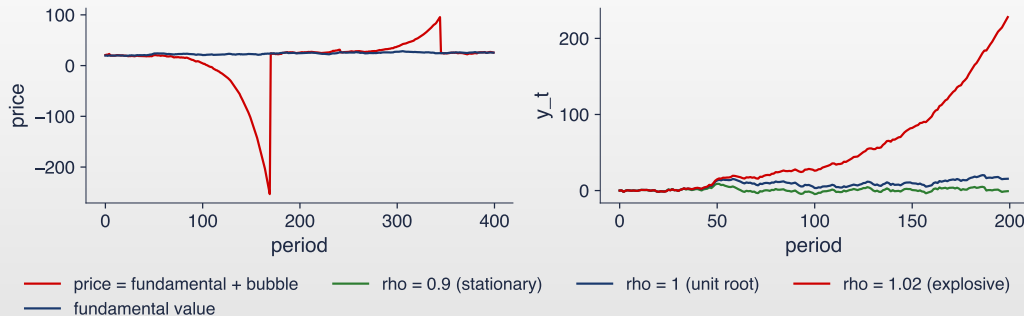
Why a rational bubble must explode

- No-arbitrage pricing: $P_t = \frac{E_t[P_{t+1} + D_{t+1}]}{1 + r}$; the fundamental value F_t is one solution
 - ▶ in words: today's price is the discounted expected value of tomorrow's price plus tomorrow's dividend
 - ▶ every $P_t = F_t + B_t$ with $E_t[B_{t+1}] = (1 + r) B_t$ is also a solution
- The bubble must grow, in expectation, at the rate r : $E_t[B_{t+k}] = (1 + r)^k B_t$
 - ▶ an investor holds an asset above its value only if the overvaluation is expected to grow
 - ▶ $1 + r > 1$: the bubble component is an **explosive** process
- Consequence: if dividends are $I(1)$ ([Chapter 3](#)), the price is $I(1)$ without a bubble and explosive with one

Worked example: a bubble that can burst

- **Blanchard and Watson (1982)**: each period the bubble survives with probability π and then grows to $\frac{1+r}{\pi} B_t$; otherwise it bursts ($B_{t+1} \approx 0$)
 - ▶ check: $E_t[B_{t+1}] = \pi \cdot \frac{1+r}{\pi} B_t + (1 - \pi) \cdot 0 = (1 + r)B_t$
- With $r = 2\%$ and $\pi = 0.98$ per period:
 - ▶ while it survives the bubble grows by $1.02/0.98 - 1 = 4.08\%$ per period, more than r : the extra growth pays for the risk of a burst
 - ▶ expected lifetime $1/(1 - \pi) = 50$ periods; it survives more than 34 periods with probability $1/2$ ($0.98^k = 0.5$)
- The longer the bubble lasts, the faster it must grow and the larger the fall when it bursts

A simulated rational bubble and three AR(1) roots



- Left: fundamental value (random walk) plus a Blanchard–Watson bubble ($r = 2\%$, $\pi = 0.98$), 400 periods; right: $y_t = \rho y_{t-1} + \varepsilon_t$ with the same shocks and $\rho = 0.9, 1, 1.02$

Interpreting the simulation

- The bubble grows, bursts, restarts: in this path it reaches 71 and bursts 2 times after a large size
 - ▶ the price inherits the explosive episodes; between them it follows the fundamental value
- AR(1) roots ([Chapter 3](#)): $\rho = 0.9$ reverts to the mean, $\rho = 1$ wanders (unit root), $\rho = 1.02$ explodes
 - ▶ with $\rho = 1.02$ a shock doubles in about 35 periods and is multiplied by 7.24 after 100
- Idea of the tests: an explosive root ($\rho > 1$) in the log price during a run-up is the statistical trace of a rational bubble

Right-tailed unit-root tests

- The regression of Chapter 3 on a window of the log price $y_t = \ln P_t$: $\Delta y_t = a + \delta y_{t-1} + \varepsilon_t$, with $\delta = \rho - 1$
 - ▶ Chapter 3 (Dickey and Fuller (1979)): H_0 : $\delta = 0$ against H_1 : $\delta < 0$ (stationary), the **left** tail
 - ▶ bubbles: H_0 : $\delta = 0$ (unit root) against H_1 : $\delta > 0$ (explosive), the **right** tail: reject for large t -statistics
- Problem: a bubble occupies only a part of the sample, and the crash after it looks like mean reversion
 - ▶ one ADF on the whole sample has almost no power: we must compute the test on many **windows**
- Notation
 - ▶ $\Delta y_t = y_t - y_{t-1}$; a : the intercept; ε_t : the error; ρ : the AR(1) coefficient of y_t
 - ▶ $r_1, r_2 \in [0, 1]$: fractions of the sample; $ADF_{r_1}^{r_2}$: the t -statistic of $\hat{\delta}$ on the window from $r_1 T$ to $r_2 T$; r_0 : the smallest window

SADF, GSADF and BSADF (1/2)

- sup: the largest value of the ADF statistic over all the windows considered
 - ▶ a single large value, in any window, is enough to signal an explosive episode
- **SADF** Phillips, Wu and Yu (2011)
 - ▶ start fixed at 0, the end r_2 moves forward
 - ▶ $\text{SADF} = \sup_{r_2 \in [r_0, 1]} \text{ADF}_0^{r_2}$
 - ▶ good for one bubble; after a first crash it loses power
- **GSADF** Phillips, Shi and Yu (2015a)
 - ▶ both the start r_1 and the end r_2 move
 - ▶ $\text{GSADF} = \sup_{r_2 \in [r_0, 1]} \sup_{r_1 \in [0, r_2 - r_0]} \text{ADF}_{r_1}^{r_2}$
 - ▶ detects several bubbles in one sample; smallest window $r_0 = 0.01 + 1.8/\sqrt{T}$

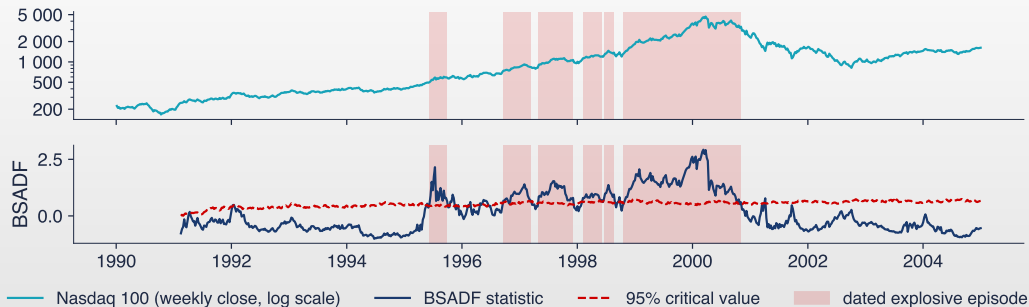
SADF, GSADF and BSADF (2/2)

- **BSADF** (date-stamping)
 - ▶ one statistic for each date r_2 , using only data up to r_2
 - ▶ $\text{BSADF}(r_2) = \sup_{r_1} \text{ADF}'_{r_1^2}$: the largest statistic over all starts r_1 of windows ending at r_2
 - ▶ an episode starts when BSADF crosses its critical value and lasts at least $\log T$ observations [Phillips, Shi and Yu \(2015b\)](#) (log: the natural logarithm)
- The distributions are not standard: critical values by Monte Carlo
 - ▶ simulated under a random walk with a weak drift, $y_t = T^{-1} + y_{t-1} + \varepsilon_t$
 - ▶ the drift T^{-1} is very small: a unit root, not an explosive root, under H_0
- Reading: an ADF, SADF or GSADF statistic above its 95% critical value rejects the unit root in favour of an explosive root

Worked example: one window of the Nasdaq 100

- Weekly log closes of the Nasdaq 100, $T = 782$ weeks; $r_0 = 0.074$, i.e. a smallest window of 58 weeks
 - ▶ end of the window: 24 March 2000, the week of the peak; among all start points the largest statistic comes from the start 17 January 1992
- OLS on this window ($n = 427$ weekly changes): $\widehat{\Delta y_t} = -0.0336 + 0.0060 y_{t-1}$
 - ▶ $t = \hat{\delta}/\text{SE}(\hat{\delta}) = 0.0060/0.0021 = 2.91$
 - ▶ so BSADF at the peak is 2.91, above its 95% critical value of 0.54
- Reading: $\hat{\delta} > 0$ means $\hat{\rho} = 1 + \hat{\delta} > 1$: the higher the price, the larger the next weekly rise, the mark of an explosive phase

Explosive episodes of the Nasdaq 100, 1990–2004

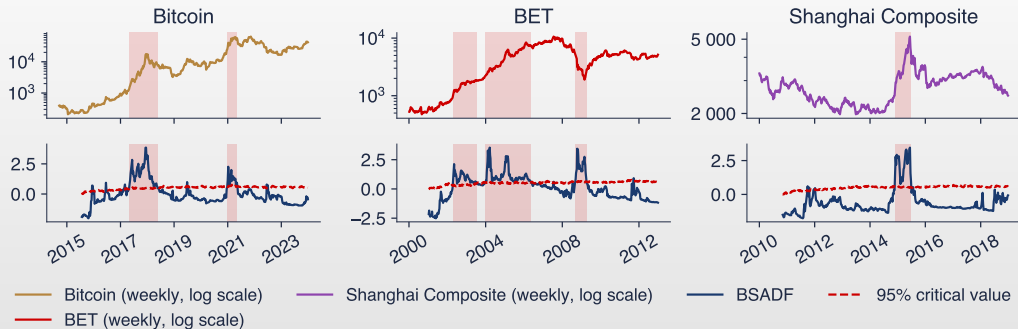


- Top: weekly close, log scale; bottom: BSADF and its 95% critical values (Monte Carlo, 1000 random walks); shaded: dated episodes (at least 7 weeks above the critical value)

Interpreting the Nasdaq 100 tests

- ▣ Whole-sample ADF -1.27 (95% critical value -0.13): no evidence
 - ▶ SADF 2.72 (critical value 1.48) and GSADF 2.92 (critical value 2.28): explosive behaviour
 - ▶ the crash of 2000–2002 hides the bubble from a single test on the whole sample
- ▣ BSADF dates 6 episodes; the longest runs from 23 October 1998 to 3 November 2000 (107 weeks)
 - ▶ short explosive bursts already from June 1995: the 1990s boom came in waves, as Phillips, Wu and Yu (2011) found
- ▣ The end date is late: the statistic stays high for months after the peak; the test dates exuberance, it does not announce the crash

Bitcoin, BET and Shanghai: dated episodes



- Weekly log closes; top: price with the dated episodes; bottom: BSADF and its 95% critical values; Bitcoin 2014–2023, BET 2000–2012, Shanghai Composite 2010–2018

Interpreting the three markets

- GSADF: Bitcoin 3.85, BET 3.53, Shanghai 3.46; all above their 95% critical values (about 2.18)
 - ▶ Bitcoin: April 2017 – May 2018 and January 2021 – May 2021; Shanghai: November 2014 – June 2015
- BET: the explosive phase is December 2003 – May 2006, not the run-up to July 2007, which was strong but not explosive
 - ▶ the BET flag of September 2008 – April 2009 is the collapse: an **accelerating fall** also gives $\hat{\delta} > 0$; always read the dates against the price chart
- The Bitcoin 2017 episode lasts until May 2018, again well after the peak

Limits of the explosive-root tests

□ Fundamentals

- ▶ an explosive price is a bubble only if the fundamentals are not explosive
- ▶ better: test the price–dividend ratio ([Phillips, Shi and Yu \(2015a\)](#) for the S&P 500); for Bitcoin there are no dividends at all

□ Volatility

- ▶ changing volatility distorts the size of the test
- ▶ a wild bootstrap gives robust critical values [Phillips and Shi \(2020\)](#)

□ Timing

- ▶ BSADF uses only past data, so it can run in real time
- ▶ but it signals a bubble that is already under way and says nothing about when it ends
- ▶ the LPPL model of the next section tries to answer exactly this: when?

Recap: Rational bubbles and explosive roots

- ▣ A rational bubble grows in expectation at the rate r : $E_t[B_{t+1}] = (1 + r)B_t$, an explosive process
- ▣ Right-tailed ADF: $H_1: \delta > 0$; SADF and GSADF take the supremum over windows; BSADF dates the episodes
- ▣ Nasdaq 100, Bitcoin, BET and Shanghai all show explosive episodes; the tests date the run-up, not the crash

Crashes as critical points



Didier Sornette (born 1957)

Photo: Didier Sornette (2012); CC BY-SA 3.0 DE;

[Wikimedia Commons](#)

- Physicist, ETH Zürich; with Anders Johansen and Olivier Ledoit he proposed the **JLS model** [Johansen, Ledoit and Sornette \(2000\)](#)
- Idea: traders imitate their neighbours; imitation creates positive feedback
 - ▶ rising prices attract buyers, who push prices higher: growth accelerates
 - ▶ the market becomes more and more fragile, like a physical system near a **critical point** (water near boiling)
- The **critical time** t_c
 - ▶ the most probable moment of the end of the bubble
 - ▶ a crash is likely near t_c but not certain: the bubble may also end with a slow decline

Super-exponential growth

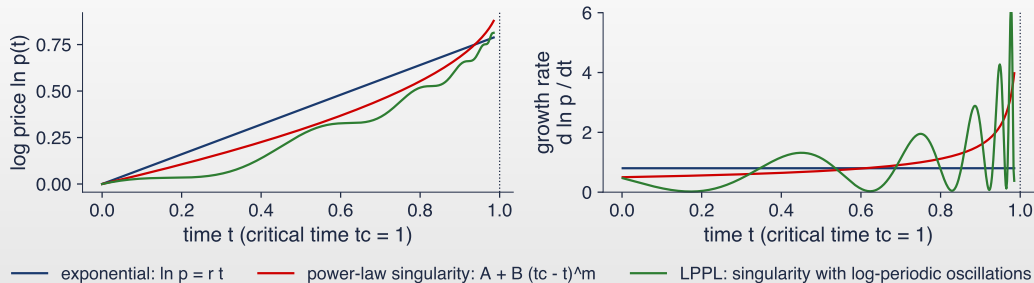
□ Exponential growth

- ▶ constant growth rate, $\ln P(t) = a + g t$
- ▶ $d \ln P / dt = g$
- ▶ a : the initial log price; g : the constant growth rate; $d \ln P / dt$: the growth rate of the price at time t
- ▶ a straight line on a log scale

□ Super-exponential growth

- ▶ the growth rate increases with time
 - ▶ on a log scale the curve bends upward
 - ▶ power-law singularity: $\ln P(t) = A + B (t_c - t)^m$, with $B < 0$ and $0 < m < 1$
 - ▶ t_c : the critical time; $t_c - t$: the time left until it; A : the log price reached at t_c ; B : the size of the growth; m : the exponent
 - ▶ growth rate $d \ln P / dt = -Bm (t_c - t)^{m-1} \rightarrow \infty$ as $t \rightarrow t_c$, while $\ln P(t_c) = A$ stays finite
- Such growth cannot last: it must end at or before t_c , a **finite-time singularity**

Exponential and super-exponential growth



- Left: log price; right: its growth rate $d \ln P / dt$; exponential ($g = 0.8$), power-law singularity ($A = 1$, $B = -1$, $m = 0.5$, $t_c = 1$) and the same with log-periodic oscillations

Interpreting the growth paths

- All three start with similar growth rates: early in a bubble the paths are hard to tell apart
- The exponential rate stays at 0.8; the singular rate rises from 0.50 to 0.70 halfway and to 3.98 just before t_c
- The oscillations ride on the accelerating trend and become faster near t_c : the next slides explain why

From a crash hazard to the LPPL equation (1/2)

- JLS: the relative price change before the crash has three parts
 - ▶ $dP/P = \mu(t) dt + \sigma dW - \kappa dj$
 - ▶ $\mu(t) dt$: the expected growth over a short interval dt ; $\mu(t)$: the drift, which may change in time
 - ▶ σdW : the ordinary noise; W : a Brownian motion (a continuous random walk); σ : the volatility
 - ▶ κdj : the crash; j jumps from 0 to 1 at the crash, which removes a fraction κ of the price
- **Hazard rate** $h(t)$
 - ▶ the probability per unit of time that the crash happens now, given that it has not happened yet

From a crash hazard to the LPPL equation (2/2)

- No arbitrage: $E[dP] = 0 \Rightarrow \mu(t) = \kappa h(t)$
 - ▶ $E[dj] = h(t) dt$, so the expected change is zero only if the drift pays for the expected crash loss
 - ▶ the higher the hazard, the faster the price must grow: risk and growth increase together
- Imitation near a critical point gives a hazard that rises towards t_c with oscillations
 - ▶ $h(t) \approx \alpha(t_c - t)^{m-1}[1 + \beta \cos(\omega \ln(t_c - t) - \phi')]$
 - ▶ $\alpha > 0$: the level of the hazard; $(t_c - t)^{m-1}$ grows without bound as $t \rightarrow t_c$, since $m < 1$
 - ▶ β : the relative size of the oscillations; ω : their log-frequency; ϕ' : their phase
- Integrating $\mu(t) = \kappa h(t)$ over time gives the log price on the next slide

The LPPL equation

- **Log-periodic power law (LPPL)**, for $t < t_c$

► $\ln P(t) = A + B (t_c - t)^m + C (t_c - t)^m \cos(\omega \ln(t_c - t) - \phi)$

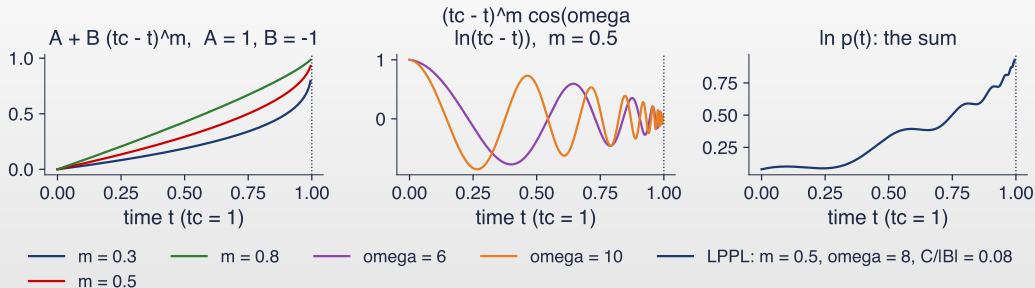
Parameter	Meaning	Usual range
t_c	critical time: most probable end of the bubble	after the last observation
A	log price at t_c (if the bubble reached t_c)	$A > 0$
B	size of the power-law growth	$B < 0$ (rising price)
m	exponent: how fast the growth accelerates	$0 < m < 1$
C	relative size of the oscillations	$ C < B $
ω	angular log-frequency of the oscillations	$2 \leq \omega \leq 25$
ϕ	phase of the oscillations	$[0, 2\pi)$

- Seven parameters: three of them (t_c , m , ω) enter nonlinearly; ϕ disappears in the estimation (Section 4)

Log-periodic oscillations

- $\cos(\omega \ln(t_c - t) - \phi)$ is periodic in $\ln(t_c - t)$, not in t : one full cycle each time $\ln(t_c - t)$ changes by $2\pi/\omega$
 - ▶ successive peaks t_n satisfy $\frac{t_c - t_n}{t_c - t_{n+1}} = \lambda = e^{2\pi/\omega}$: the time left to t_c shrinks by the same factor at each cycle
- **Discrete scale invariance**
 - ▶ the pattern looks the same when time to t_c is rescaled by λ (as in a hierarchy of traders, groups, institutions)
- Worked example: with $\omega = 8$, $\lambda = e^{2\pi/8} = 2.19$
 - ▶ if a correction happens 200 days before t_c , the next one comes about $200/2.19 \approx 91$ days before t_c , then about 42, then 19
 - ▶ $\omega = 6.28$ gives $\lambda = 2.72$; $\omega = 10$ gives $\lambda = 1.87$; the empirical literature reports λ near 2 [Sornette \(2017\)](#)

The pieces of the LPPL equation



- Left: $A + B(t_c - t)^m$ for $m = 0.3, 0.5, 0.8$; middle: the oscillation $(t_c - t)^m \cos(\omega \ln(t_c - t))$ for $\omega = 6$ and 10 ; right: the full LPPL path; $t_c = 1$

Interpreting the components

- A smaller m concentrates the growth close to t_c : the curve is flat for a long time, then shoots up
 - ▶ m close to 1 is almost a straight line: hardly a bubble; m close to 0 is a jump at t_c
- The oscillation term shrinks with $(t_c - t)^m$ and speeds up: the cycles get shorter by the factor λ
 - ▶ a larger ω gives more cycles before t_c
- Real run-ups (Section 4) show this mixture: an accelerating trend with corrections that come more and more often

Worked example: an LPPL value by hand

- Parameters: $A = 8$, $B = -1.2$, $C = 0.08$, $m = 0.5$, $\omega = 8$, $\phi = 0$, $t_c = 2$ (years); compute $\ln P(t)$
 - ▶ $t = 1$: $t_c - t = 1$, $(t_c - t)^m = 1$, $\ln 1 = 0$, $\cos 0 = 1$: $\ln P = 8 - 1.2 + 0.08 = 6.880$, $P = 973$
 - ▶ $t = 1.75$: $(0.25)^{0.5} = 0.500$, $\ln 0.25 = -1.386$, $\cos(8 \cdot (-1.386)) = 0.095$:
 $\ln P = 8 - 1.2 \cdot 0.500 + 0.08 \cdot 0.500 \cdot 0.095 = 7.404$, $P = 1642$
 - ▶ $t = 1.99$: $(0.01)^{0.5} = 0.100$, $\cos(8 \cdot (-4.605)) = 0.654$: $\ln P = 7.885$, $P = 2658$
- The price rises from 973 to 2658 as t goes from 1 to 1.99 and would reach $e^A = 2981$ at t_c ; most of the rise comes in the last months

Parameter constraints: the filter

- A fit is **qualified** only if it describes a credible bubble (Shu and Zhu (2020), following Sornette et al. (2015)):
 - ▶ $B < 0$: rising price; $0.01 \leq m \leq 0.99$: accelerating growth with a finite price at t_c
 - ▶ $2 \leq \omega \leq 25$: neither too slow nor too fast oscillations; at least 2.5 oscillations in the window, $\frac{\omega}{\pi} \ln \frac{t_c - t_1}{t_c - t_2} \geq 2.5$
 - ▶ $t_2 \leq t_c \leq t_2 + 0.2(t_2 - t_1)$: the end is close, relative to the length of the window $[t_1, t_2]$
 - ▶ damping $\frac{m|B|}{\omega|C|} \geq 1$: the hazard rate stays positive
- Conditions on the residuals
 - ▶ every fitted price within 15% of the observed price; the Lomb periodogram Lomb (1976) confirms the log-periodic cycle (level 10%)
 - ▶ Lomb periodogram: a periodogram for unevenly spaced points, here of the residuals as a function of $\ln(t_c - t)$
 - ▶ the residuals are stationary: the Dickey–Fuller and Phillips–Perron tests of Chapter 3 reject a unit root (level 10%)

Recap: The LPPL model

- Imitation creates a hazard rate that grows towards t_c ; no arbitrage turns it into super-exponential growth
- $\ln P(t) = A + B(t_c - t)^m + C(t_c - t)^m \cos(\omega \ln(t_c - t) - \phi)$, seven parameters
- Log-periodic: cycles shrink by $\lambda = e^{2\pi/\omega}$; a fit counts only if it passes the filter

The two-step method of Filimonov and Sornette (1/2)

- Least squares over seven parameters has many local minima; Filimonov and Sornette (2013) reduce the search to three parameters
 - ▶ $\min \text{SSR} = \sum_{t=t_1}^{t_2} [\ln P_t - \text{LPPL}(t)]^2$; SSR: the sum of squared residuals over the window $[t_1, t_2]$
 - ▶ $\text{LPPL}(t)$: the right-hand side of the LPPL equation
- Expand the cosine, with $\tau = t_c - t$
 - ▶ $C \cos(\omega \ln \tau - \phi) = C_1 \cos(\omega \ln \tau) + C_2 \sin(\omega \ln \tau)$, where $C_1 = C \cos \phi$, $C_2 = C \sin \phi$
- The LPPL equation becomes linear in A, B, C_1, C_2
 - ▶ $\ln P(t) = A + B f_t + C_1 g_t + C_2 h_t$
 - ▶ $f_t = \tau^m$, $g_t = \tau^m \cos(\omega \ln \tau)$, $h_t = \tau^m \sin(\omega \ln \tau)$: known numbers once t_c, m, ω are fixed

The two-step method of Filimonov and Sornette (2/2)

□ Step 1 (linear)

- ▶ for given (t_c, m, ω) , regress $\ln P_t$ on $(1, f_t, g_t, h_t)$ by OLS
- ▶ $(\hat{A}, \hat{B}, \hat{C}_1, \hat{C}_2) = (X'X)^{-1}X'y$; X : the matrix with columns $1, f_t, g_t, h_t$; y : the vector of $\ln P_t$
- ▶ this gives the concentrated cost $SSR(t_c, m, \omega)$, a function of three parameters only

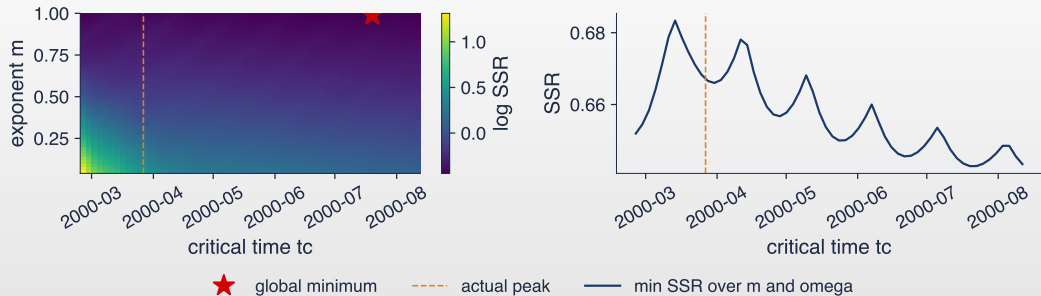
□ Step 2 (nonlinear)

- ▶ minimise $SSR(t_c, m, \omega)$ over the search space
- ▶ $t_c \in [t_2, t_2 + (t_2 - t_1)/3]$, $m \in [0, 1]$, $\omega \in [1, 50]$
- ▶ in the code: a grid of $8 \times 8 \times 16$ points, then the Nelder–Mead simplex from the best grid point

□ At the end: $C = \sqrt{C_1^2 + C_2^2}$ and $\phi = \text{atan2}(C_2, C_1)$

- ▶ $\text{atan2}(C_2, C_1)$: the angle of the point (C_1, C_2) , in $(-\pi, \pi]$

The cost landscape



□ Nasdaq 100, window from the low of October 1998 to 30 days before the peak

- ▶ left: log SSR over (t_c, m) , minimised over $\omega \in [2, 25]$ (135 360 grid points, with the linear step solved at each point)
- ▶ right: the minimum over m and ω as a function of t_c

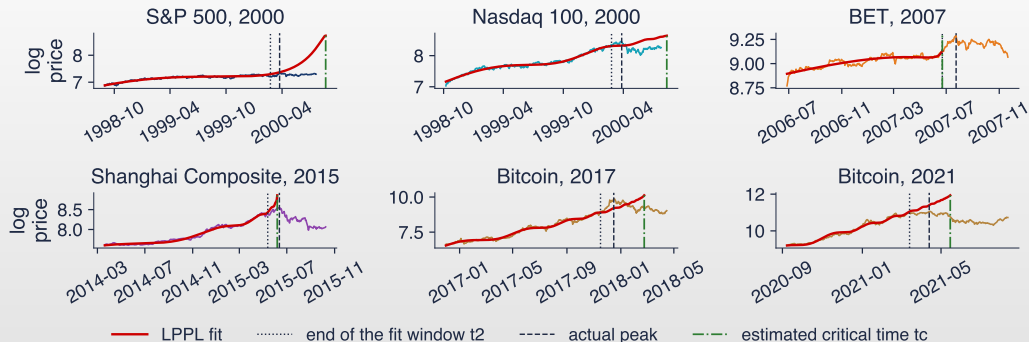
Interpreting the cost landscape

- The profile over t_c has 5 local minima: an optimiser started at a different t_c can end in a different valley
 - ▶ the waves come from the oscillation term: shifting t_c shifts the phase of the cycles
- The landscape is flat: the worst and the best t_c differ by only 6.3% in SSR
 - ▶ the global minimum is at m near 1 and $t_c = 19$ July 2000, far after the actual peak: the data barely identify t_c
- Lesson: one LPPL fit is never enough; we need many windows and the filter (Section 5)

Worked example: the Shanghai fit at the end of the window

- Shanghai Composite, window 20 March 2014 – 13 May 2015 (281 days); estimates: $t_c = 2015.428$ (6 June 2015), $m = 0.392$, $\omega = 5.50$
 - ▶ step 1 gives $\hat{A} = 8.893$, $\hat{B} = -1.270$, $\hat{C}_1 = 0.020$, $\hat{C}_2 = 0.074$, so $\hat{C} = 0.076$
- At $t_2 = 2015.361$ (years): $\tau = t_c - t_2 = 0.067$, $\tau^m = 0.347$, $\ln \tau = -2.702$
 - ▶ $\cos(\omega \ln \tau) = -0.656$, $\sin(\omega \ln \tau) = -0.754$
 - ▶ $\widehat{\ln P} = 8.893 + (-0.440) + (-0.024) = 8.429$; observed $\ln P = 8.384$
- In levels: fitted 4579 against observed 4376, an error of 4.7%; the three terms are the level A , the power-law growth and the oscillation

LPPL fits 30 days before the peak



- Window: from the low before the run-up to 30 calendar days before the peak; the fitted path is extended to the estimated t_c ; data after the window are shown only for comparison

Interpreting the six fits

Episode	t_2	$\hat{t}_c$	Days to peak	$\hat{m}$	$\hat{\omega}$	Param.	All
S&P 500, 2000	23 February 2000	21 August 2000	+150	1.00	1.0	no	no
Nasdaq 100, 2000	25 February 2000	11 August 2000	+137	1.00	5.9	no	no
BET, 2007	22 June 2007	22 June 2007	-32	0.69	1.0	no	no
Shanghai Composite, 2015	13 May 2015	6 June 2015	-6	0.39	5.5	yes	yes
Bitcoin, 2017	16 November 2017	22 February 2018	+68	0.71	14.1	no	no
Bitcoin, 2021	14 March 2021	15 May 2021	+32	0.82	17.4	no	no

- Days to peak: $\hat{t}_c$ minus the actual peak; Param.: the fit passes the parameter conditions; All: it also passes the residual conditions
- Only Shanghai passes the whole filter, with $\hat{t}_c$ within a week of the peak; S&P 500 and Nasdaq 100 hit the boundary $m = 1$: no super-exponential growth in these windows
- Bitcoin fails the 15% error condition: its daily swings are too large for one smooth path

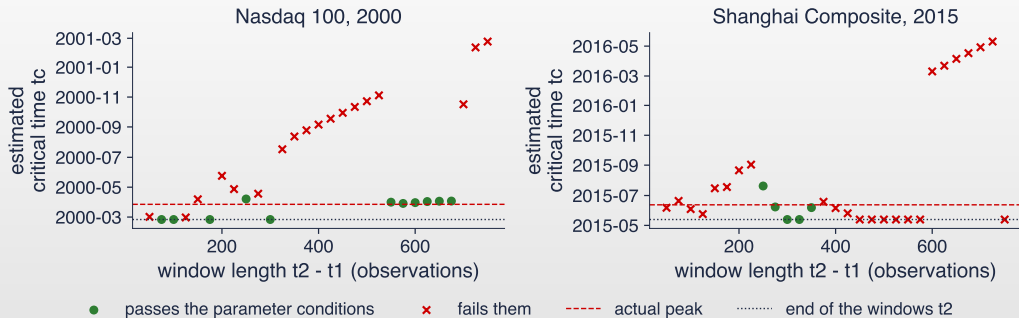
Recap: Estimation

- ▣ Four linear parameters by OLS, three nonlinear ones (t_c , m , ω) by search: the method of [Filimonov and Sornette \(2013\)](#)
- ▣ The cost surface is rugged and flat in t_c : the critical time is weakly identified
- ▣ One window, one fit: only one of six episodes gives a fully qualified fit 30 days before the peak

The LPPLS confidence indicator

- So far t_1 was the low before the run-up, a date known only **after** the bubble; in real time the start is unknown
 - ▶ remedy: for each end date t_2 (today), fit LPPL on 29 windows $[t_2 - L, t_2]$, $L = 750, 725, \dots, 50$ trading days [Shu and Zhu \(2020\)](#)
- Indicator = share of the windows ending at t_2 whose fit is qualified [Sornette et al. \(2015\)](#); 0: no window sees a bubble, 1: all do
 - ▶ LPPLS (LPPL singularity) is the name used for this indicator; the equation is the same
- Two versions in this chapter
 - ▶ **parameter conditions**: B, m, ω, t_c , oscillations, damping
 - ▶ **all conditions**: also the 15% error, the Lomb test and stationary residuals ([Shu and Zhu \(2020\)](#))
- Computed only with data up to t_2 : a real-time indicator; an alarm when it exceeds a threshold fixed in advance (here 0.2)

Many windows, one end date

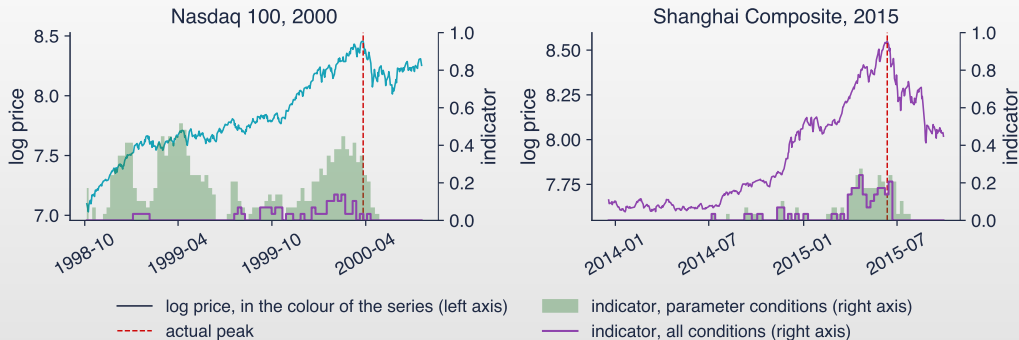


- Estimated t_c for each of the 29 windows ending 30 days before the peak; green dots: the fit passes the parameter conditions; red crosses: it fails them

Interpreting the windows

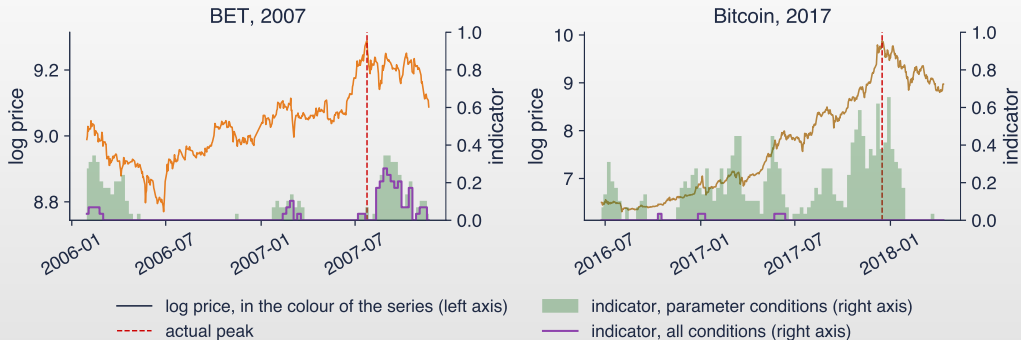
- ▣ Nasdaq 100 (end 25 February 2000): 11 of 29 windows pass the parameter conditions, 1 the whole filter
 - ▶ qualified $\hat{t}_c$: median 30 March 2000, 10%–90% range 25 February 2000 – 2 April 2000; actual peak 27 March 2000
- ▣ Shanghai (end 13 May 2015): 5 of 29 pass the parameter conditions; median $\hat{t}_c$ 6 June 2015, actual peak 12 June 2015
 - ▶ the condition on m fails most often: only 41% of the Shanghai windows have $0.01 \leq m \leq 0.99$
- ▣ Short windows put t_c right after t_2 ; long windows spread it out: the window length is a hidden modelling choice

The indicator around four peaks (1/2)



- Log price (left axis) and the indicator (right axis) every 5 trading days, from 18 months before to 4 months after the peak
- green area: parameter conditions; purple line: all conditions

The indicator around four peaks (2/2)

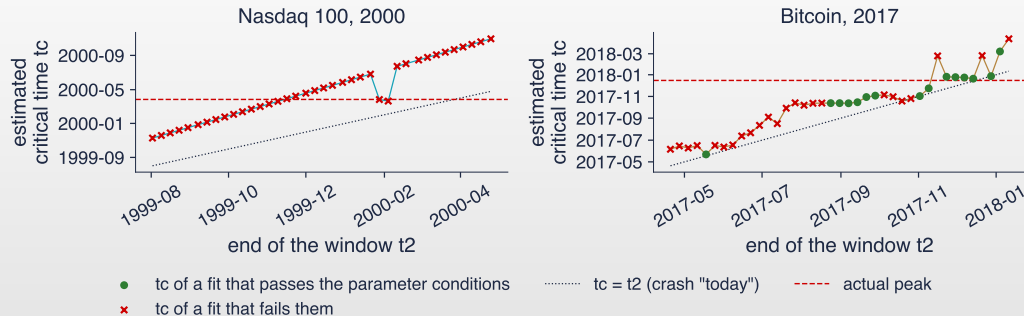


- The same layout for the BET (2007) and Bitcoin (2017); Bitcoin: one value every 7 days
- green area: parameter conditions; purple line: all conditions

Interpreting the indicator

- Nasdaq 100: the parameter version reaches 0.52 on 7 April 1999, a year before the peak, and rises again in the last months; the full version stays below 0.14
 - ▶ Shanghai: both versions rise in the two months before the peak (maximum 0.24 on 22 April 2015)
- BET: the strongest full-filter signal, 0.28, comes on 28 August 2007, after the peak: the fits then describe the last part of the run-up, not the future
 - ▶ Bitcoin 2017: the parameter version peaks at 0.66 on 29 December 2017; the full version is almost always 0 (the 15% error condition)
- The indicator does light up in bubbles, but early, late and with gaps; the choice of filter changes the picture

The critical time moves with the data



- Window start fixed at the low before the run-up; the end t_2 moves from 8 months before to 1 month after the peak (weekly); each point: the $\hat{t}_c$ of the fit ending at t_2

Interpreting the moving critical time

- Nasdaq 100: over 38 end dates $\hat{t}_c$ ranges from 9 November 1999 to 30 October 2000 (356 days); none of these fits passes the parameter conditions
 - ▶ Bitcoin 2017: range 21 May 2017 – 12 April 2018; 38% of the fits pass; median error 65 days
- Many points lie close to the line $t_c = t_2$: the fit often says “the end is now”, week after week
 - ▶ a forecast that keeps moving with the data is a description of the past, not a prediction
- Picking, after the crash, the one window whose $\hat{t}_c$ was right is **look-ahead bias**

Recap: Confidence from many windows

- ▣ The start of a bubble is unknown: fit 29 windows for each end date
- ▣ Indicator = share of qualified windows; it depends strongly on the filter
- ▣ $\hat{t}_c$ moves with the end of the data: report its spread over windows, never one value

Traps in evaluating crash predictions

- **Look-ahead bias**

- ▶ using information from after t_2
- ▶ choosing t_1 at the low of the run-up, the episodes after the crash, or the filter after seeing the results

- **Selection**

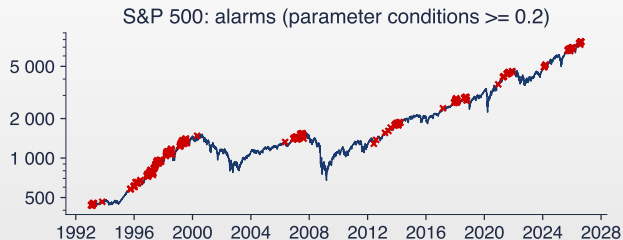
- ▶ studying only the bubbles that burst
- ▶ the run-ups that did not end in a crash are forgotten

- **False alarms and base rates**

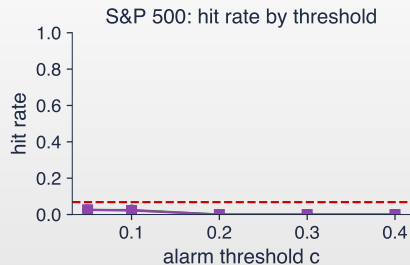
- ▶ an alarm is useful only if a crash is **more likely after an alarm** than on an ordinary day
- ▶ the comparison: $P(\text{crash} \mid \text{alarm})$ against $P(\text{crash})$, the unconditional frequency

- A fair test fixes everything in advance (windows, filter, threshold, the definition of a crash) and runs over the whole sample

Alarms over the whole sample (1/2)



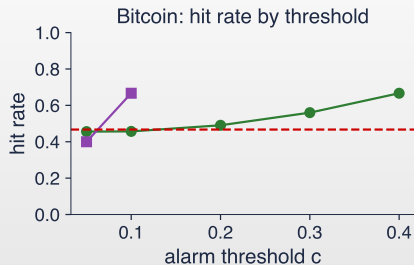
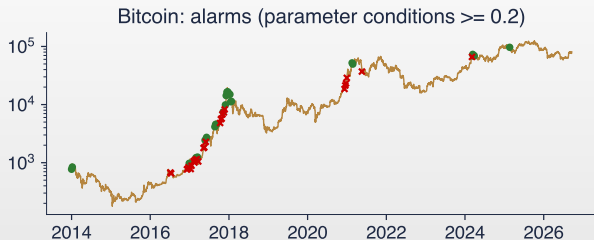
— S&P 500
 • alarm followed by a fall of 20% or more
 × false alarm
 — hit rate, parameter conditions



— hit rate, all conditions
 - - - unconditional frequency

- S&P 500: indicator every 5 trading days, January 1993 – September 2026
- ▶ a crash: a fall of at least 20% within 182 days after the alarm; right: hit rate by threshold

Alarms over the whole sample (2/2)



- Bitcoin
- alarm followed by a fall of 20% or more
- × false alarm
- hit rate, parameter conditions
- hit rate, all conditions
- - - unconditional frequency

Bitcoin: indicator every 7 days, January 2014 – September 2026

► hit rate: the share of the alarm dates followed by a crash; dashed line: the unconditional frequency

Interpreting the evaluation

- S&P 500: a fall of 20% within six months follows 7% of all dates, but 0% of the alarm dates (threshold 0.2, parameter conditions)
 - ▶ 20 alarm clusters, none followed by such a fall within six months; the alarms come in calm, steady uptrends
 - ▶ of the 4 drawdowns of 20% or more since January 1993, 2 had an alarm in the six months before their peak (2007 and 2022), but the falls came later
- Bitcoin: base rate 47%, after an alarm 49%: almost no gain; 12 of its 14 drawdowns of 20% had an alarm before the peak
 - ▶ falls of 20% are so frequent in Bitcoin that almost any alarm looks “right”; higher thresholds help a little, on very few dates (56% at 0.3, on 4% of the dates)
- The indicator says “this looks like a bubble regime”, not “the crash comes within six months”

The literature on crash prediction

- ▣ Successes reported by the authors of the method
 - ▶ Jiang et al. (2010): the Chinese bubbles of 2007 and 2009 diagnosed in advance; Sornette et al. (2015): the Shanghai 2015 peak announced before it happened
 - ▶ Gerlach, Demos and Sornette (2019): Bitcoin's bubbles of 2012–2018 dissected with the confidence indicator
- ▣ Critical evaluations
 - ▶ Feigenbaum (2001): the log-periodic term adds little once the estimation error is taken into account
 - ▶ Brée and Joseph (2013): on many crashes, the LPPL conditions are rarely all met before the crash; parameters are unstable
- ▣ Our own numbers point the same way: the indicator reacts to strong run-ups, but its timing is loose and its alarms are often false

Strengths and limits of LPPL

▣ Can

- ▶ describe a run-up with accelerating growth and shrinking corrections in a few interpretable parameters
- ▶ measure “how bubble-like” a market is today, across many windows (the indicator)

▣ Cannot

- ▶ give the date of a crash: t_c marks the end of the regime
 - ▶ guarantee a crash: the regime may end with a slow decline
 - ▶ tell whether the price is above the fundamental value: that needs fundamentals
- ▣ Use it with the explosive-root tests, with fundamentals (valuation ratios) and with risk measures such as VaR 1% ([Chapter 5](#)), not alone

Recap: An honest evaluation

- ▣ Fix windows, filter, threshold and the definition of a crash before looking at the outcomes
- ▣ Compare the hit rate after alarms with the base rate; count the false alarms
- ▣ LPPL describes bubbles well; as a crash-timing tool its record is weak

Possible contribution of AI

▣ Code

- ▶ a first version of the BSADF recursion, of the Filimonov–Sornette two-step fit, of the indicator over many windows

▣ Explanation

- ▶ a second explanation of the hazard-rate argument or of log-periodicity, with your own numbers

▣ Exploration

- ▶ the indicator on many markets and periods, with different filters, under a protocol fixed in advance

▣ Example prompt

- ▶ Write Python code that fits the LPPL model to the daily log price of the Shanghai Composite between two dates with the Filimonov-Sornette method (OLS for A, B, C1, C2; grid search plus Nelder-Mead for t_c , m , ω), checks the filter conditions of Shu and Zhu (2020) and plots the fit.

Checks you must run

- Simulate an LPPL path with known parameters plus noise and check that the code recovers t_c , m and ω
- The sign conventions: $B < 0$ for a rising bubble; the test of explosive roots uses the right tail, not the left tail of [Chapter 3](#)
- Critical values: SADF and GSADF need their own simulated critical values, not the Dickey–Fuller table
- No look-ahead: every quantity at date t_2 must use only data up to t_2
- A crash “predicted” by an AI answer after the fact is not a prediction; ask for the full list of alarms and false alarms
- Every cited reference: it must exist; check the DOI

Key takeaways

- ▣ A bubble is a price above its fundamental value; a rational bubble must grow explosively
- ▣ Right-tailed ADF tests (SADF, GSADF, BSADF) detect and date explosive episodes with data up to each date
- ▣ LPPL: super-exponential growth with log-periodic oscillations ending at a critical time t_c
- ▣ Estimate in two steps (OLS + search over t_c, m, ω), filter the fits, use many windows
- ▣ Judge predictions by hit rates against base rates, with all the false alarms: the timing of crashes remains hard

Key formulas

Quantity	Formula
Rational bubble	$P_t = F_t + B_t, \quad E_t[B_{t+1}] = (1+r)B_t$
Right-tailed ADF	$\Delta y_t = a + \delta y_{t-1} + \varepsilon_t, \quad H_1: \delta > 0$
GSADF, BSADF	$\sup_{r_2} \sup_{r_1} \text{ADF}_{r_1}^2, \quad \text{BSADF}(r_2) = \sup_{r_1} \text{ADF}_{r_1}^2, \quad r_0 = 0.01 + 1.8/\sqrt{T}$
LPPL	$\ln P(t) = A + B(t_c - t)^m + C(t_c - t)^m \cos(\omega \ln(t_c - t) - \phi)$
Linear form	$\ln P(t) = A + Bf_t + C_1 g_t + C_2 h_t, \quad C = \sqrt{C_1^2 + C_2^2}$
Scaling ratio	$\lambda = e^{2\pi/\omega}$
Damping	$m B /(\omega C) \geq 1$
Indicator	share of qualified windows ending at t_2

Self-assessment (1/2)

- **Question:** in a Blanchard–Watson bubble with $r = 1\%$ and $\pi = 0.95$, by how much does the bubble grow in a period in which it survives?
 - ▶ **Answer:** by $1.01/0.95 - 1 \approx 6.3\%$; expected lifetime $1/(1 - 0.95) = 20$ periods
- **Question:** a right-tailed ADF on a window gives $\hat{\delta} = 0.004$ with SE 0.002. Is the window explosive at 5% if the critical value is 1.5?
 - ▶ **Answer:** yes: $t = 0.004/0.002 = 2 > 1.5$; we reject the unit root in favour of an explosive root
- **Question:** why do we use the right tail and not the left tail of [Chapter 3](#)?
 - ▶ **Answer:** the alternative is $\rho > 1$ (explosive), not $\rho < 1$ (stationary); evidence for it is a large positive t

Self-assessment (2/2)

- **Question:** an LPPL fit gives $\omega = 6.28$. By what factor does the time left to t_c shrink between two corrections?
▶ **Answer:** $\lambda = e^{2\pi/6.28} \approx e \approx 2.72$
- **Question:** a fit has $m = 1.0$ at the edge of the search space. Is it a qualified bubble fit?
▶ **Answer:** no: the filter needs $m \leq 0.99$; $m = 1$ is a linear trend in t , not an accelerating one
- **Question:** an indicator gives alarms before 8 of 10 crashes. Is it a good predictor?
▶ **Answer:** not yet: we also need the false alarms and the base rate; if alarms are on almost every day, catching 8 crashes is easy

Project idea

- **Is the BET in a bubble today?** A real-time protocol, fixed before looking at the result
 - ▶ BSADF on the weekly BET and BET-TR; the indicator of this chapter on the daily BET
 - ▶ the evaluation of Section 6 on the BET since 2000: alarms, false alarms, base rate
- Extensions
 - ▶ the price–earnings ratio instead of the price; Bitcoin and Ethereum together
 - ▶ compare with the regime-switching models of [Chapter 10](#)
- Further reading on the Bucharest Stock Exchange: [Pele and Mazurencu-Marinescu \(2012\)](#)
- Next: [Chapter 14](#), multivariate GARCH models

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